

| Question | Answer                                                                                | Marks     |
|----------|---------------------------------------------------------------------------------------|-----------|
| 3(a)(i)  | e.g. period = $6 / 2.5$                                                               | <b>C1</b> |
|          | frequency = $0.42 \text{ Hz}$                                                         | <b>A1</b> |
| 3(a)(ii) | energy = $\frac{1}{2} m \times 4\pi^2 f^2 y_0^2$                                      | <b>C1</b> |
|          | = $\frac{1}{2} \times 0.25 \times 4\pi^2 \times 0.42^2 \times (1.5 \times 10^{-2})^2$ | <b>C1</b> |
|          | = $2.0 \times 10^{-4} \text{ J}$                                                      | <b>A1</b> |
| 3(b)(i)  | (induced) e.m.f. proportional to rate of                                              | <b>M1</b> |
|          | change of magnetic flux (linkage)<br><b>or</b><br>cutting of magnetic flux            | <b>A1</b> |
| 3(b)(ii) | coil cuts flux/field (of moving magnet) <u>inducing</u> e.m.f. in coil                | <b>B1</b> |
|          | (induced) current in resistor causes heating (effect)                                 | <b>M1</b> |
|          | thermal energy/heat derived from energy of oscillations (of magnet)                   | <b>A1</b> |

9702/42

Cambridge International AS/A Level – Mark Scheme

May/June 2017

**PUBLISHED**

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| 4(c)     | rules (of ultrasound) | B1    |